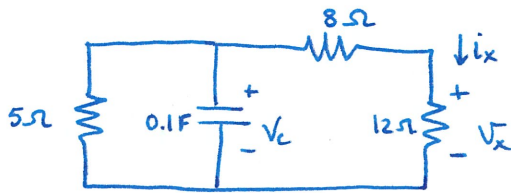


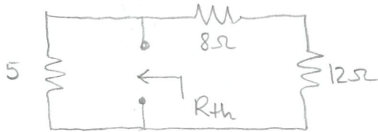
EXERCISE 3 (Natural response RC)



Calculate V_c , V_x and i_x if $V_c(0) = 15V$

Natural response

First, calculate time constant $\tau = R_{th}C$



$$R_{th} = 5 \parallel (12+8) = 5 \parallel 20 = 4\Omega$$

$$\frac{1}{\frac{1}{5} + \frac{1}{20}} = \frac{1}{\frac{4+1}{20}} = \frac{20}{5} = 4$$

$$\tau = 4\Omega \times 0.1F = 0.4s$$

Then, use natural response formula

$$V(t) = V_0 e^{-t/\tau} = 15 e^{-t/0.4} = \underline{\underline{15 e^{-2.5t} V}}$$

We use voltage division to find V_x

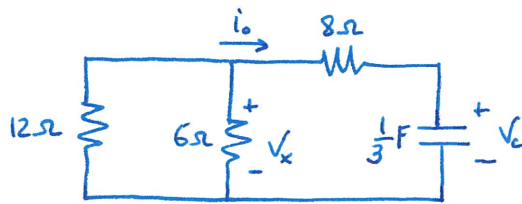
$$V_x = V_c \frac{12}{12+8} = \frac{12}{20} V_c = \frac{3}{5} \times 15 e^{-2.5t} = \underline{\underline{9 e^{-2.5t}}}$$

We use Ohm's law to calculate i_x

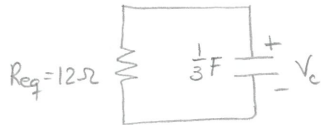
$$i_x = \frac{V_x}{12} = \frac{9}{12} e^{-2.5t} = \underline{\underline{\frac{3}{4} e^{-2.5t}}}$$

EXERCISE 4 (Natural response RC)

Find V_c , V_x and i_o if $V_c(0) = 60\text{ V}$



Simplify the circuit :



$$R_{eq} = (12 \parallel 6) + 8 = \frac{1}{\frac{1}{12} + \frac{1}{6}} + 8 = \frac{1}{\frac{3}{12}} + 8 = \frac{12}{3} + 8 = 12\Omega \Rightarrow \tau = RC = 4\text{ s}$$

Use natural response formula

$$V_c(t) = V_0 e^{-t/\tau} = \underline{\underline{60 e^{-t/4} \text{ V}}}$$

Use voltage division to find V_x :

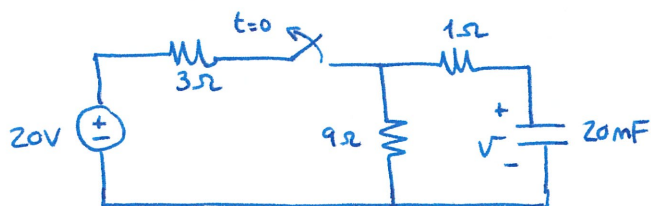
$$V_x = V_c \frac{6 \parallel 12}{8 + 6 \parallel 12} = V_c \frac{4}{12} = \frac{1}{3} V_c = \frac{1}{3} 60 e^{-t/4} = \underline{\underline{20 e^{-t/4} \text{ V}}}$$

Use $i_c = C \frac{dV_c}{dt}$ to calculate i_o :

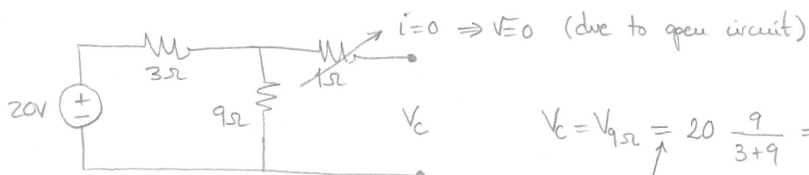
$$i_o = i_c = C \frac{dV_c}{dt} = \frac{1}{3} \frac{d}{dt} (60 e^{-t/4}) = \frac{60}{3} \left(-\frac{1}{4}\right) e^{-t/4} = -\frac{60}{12} e^{-t/4} = \underline{\underline{-5 e^{-t/4} \text{ A}}}$$

EXERCISE 5 (Natural response RC)

Find initial energy stored in the capacitor and $v(t)$ for $t \geq 0$.



* $t \leq 0 \Rightarrow$ DC steady state $\Rightarrow C \equiv$ open circuit

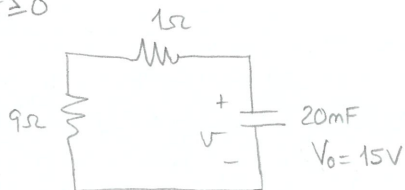


$$V_c = V_{9\Omega} = 20 \frac{9}{3+9} = \frac{180}{12} = 15 \text{ V} = V_0$$

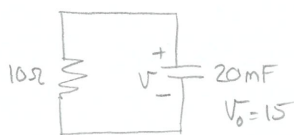
Voltage divider

$$\underline{W_c(0) = \frac{1}{2} C V_c^2(0) = \frac{1}{2} C V_0^2 = \frac{1}{2} 20 \times 10^{-3} \times 15^2 = 2.25 \text{ J}}$$

* $t \geq 0$



simplify



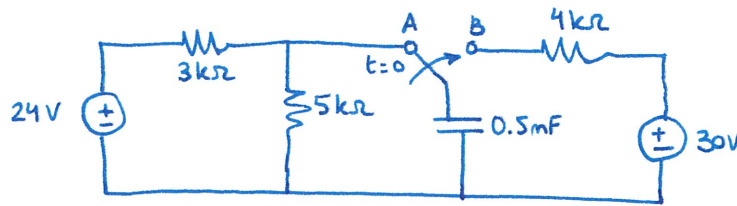
Use natural response formula:

$$\tau = RC = 10\Omega \times 20\text{mF} = 0.2\text{s}$$

$$v(t) = V_0 e^{-t/\tau} = 15 e^{-t/0.2} = \underline{\underline{15 e^{-5t} \text{ V}}}$$

EXERCISE 7 (Step response RC)

Calculate $v(t)$ for $t > 0$ and calculate its value at $t = 1s$ and $t = 4s$.



Calculate V_0 first, for $t < 0$



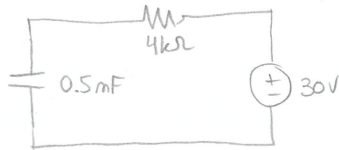
DC steady state
 $C \equiv$ open circuit



$$V_0 = V_{5k} = 24 \frac{5k}{5+3k} = \frac{24 \times 5}{8} = 15V$$

voltage divider

Calculate $V(t)$ for $t > 0$



$$\tau = RC = 4k\Omega \times 0.5mF = 2s$$

(since there is only one resistor, $R = R_{th}$)

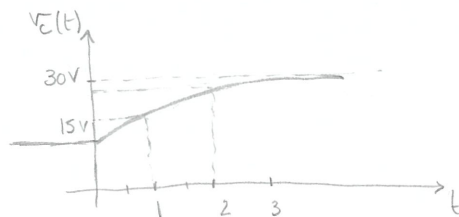
Use step response formula:

$$V(t) = \underset{V(\infty)}{V_s} + \underset{V(\infty)}{(V_0 - V_s)} e^{-t/\tau} = 30 + (15 - 30)e^{-t/2} = \underline{\underline{30 - 15e^{-0.5t}}}$$

Calculate $V(1)$ and $V(4)$:

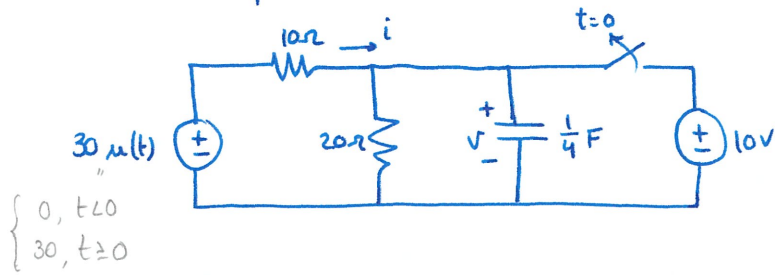
$$V(1) = 30 - 15e^{-0.5} = 20.9V$$

$$V(4) = 30 - 15e^{-0.5 \times 4} = 30 - 15e^{-2} = 27.97V$$

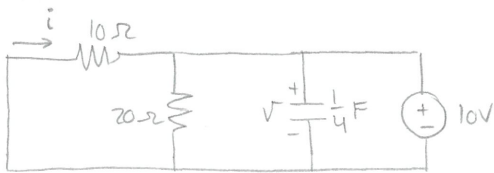


EXERCISE 8 (Step response RC)

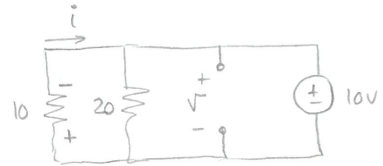
Find v and i for all time



* $t < 0$



DC steady state
 $C \equiv$ open circuit



$$v = 10V, t < 0 \Rightarrow V_0 = 10V$$

$$i = \frac{-v}{10} = \frac{-10V}{10\Omega} = -1A$$

Ohm's law
voltages in parallel, but with different polarities

* $t \geq 0$



Calculate Thevenin equivalent before using step response formula

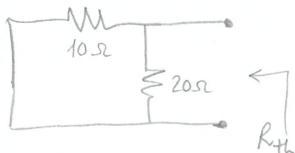
$$\rightarrow V_{th} = V_{oc} = V_{oc}$$



$$V_{th} = 30 \frac{20}{20+10} = \frac{600}{30} = 20V$$

voltage divider

$$\rightarrow R_{th}:$$



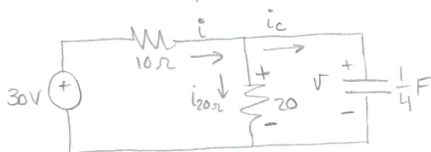
$$R_{th} = 20 \parallel 10 = \frac{1}{\frac{1}{20} + \frac{1}{10}} = \frac{1}{\frac{1+2}{20}} = \frac{20}{3} \Omega$$

$$\tau = R_{th} \times C = \frac{20}{3} \Omega \times \frac{1}{4} F = 5/3 s$$

Use step response formula

$$v(t) = V_{oc} + (V_0 - V_{oc}) e^{-t/\tau} = 20 + (10 - 20) e^{-t/5/3} = 20 - 10 e^{-3t/5} V$$

Use KCL to find current i :



$$i = i_{20\Omega} + i_C$$

$$i(t) = \frac{v(t)}{20} + C \frac{dv(t)}{dt} = \frac{1}{20} (20 -$$

$$10 e^{-3t/5}) + \frac{1}{4} (-10) \left(-\frac{3}{5}\right) e^{-3t/5}$$

$$i(t) = 1 - 0.5 e^{-3t/5} + \frac{3}{2} e^{-3t/5}$$

$$i(t) = 1 + e^{-3t/5}$$